BGP Interview Questions and Answers - Expanded

Q1. What is BGP?

Border Gateway Protocol is the Internet's path-vector routing protocol. It exchanges routing information between Autonomous Systems (AS). BGP makes routing decisions based on attributes and policies rather than only shortest paths. It is highly scalable and used by ISPs and enterprises. BGP runs over TCP port 179 for reliable communication. It is considered the de facto routing protocol of the Internet.

Q2. What is an Autonomous System?

An AS is a collection of networks under a single administrative control. Each AS is identified by a unique ASN. BGP uses AS numbers to exchange routes between organizations. Examples include ISPs, cloud providers, and enterprises. Public ASNs are used on the Internet while private ASNs are used internally. ASNs can be 2-byte or 4-byte values.

Q3. Difference between eBGP and iBGP?

eBGP runs between different Autonomous Systems. iBGP runs within the same Autonomous System. eBGP neighbors are typically directly connected while iBGP is used to distribute external routes internally. Both use TCP port 179. eBGP routes are generally preferred over iBGP routes. iBGP requires full mesh or route reflectors for scalability.

Q4. Why is BGP called a Path Vector Protocol?

BGP advertises the complete AS Path to a destination. Routers analyze this path before selecting routes. The AS Path also helps prevent routing loops. This makes BGP different from distance-vector protocols. Multiple attributes are carried with routes. Policy-based routing decisions are a key feature of BGP.

Q5. What is the default Administrative Distance of BGP?

eBGP has an Administrative Distance of 20. iBGP has an Administrative Distance of 200. This allows external BGP routes to be preferred over many internal routing protocols. Administrative Distance determines route trustworthiness. Lower values are preferred by the routing table.

Q6. What is AS Path?

AS Path is a list of Autonomous Systems a route has traversed. BGP uses it for loop prevention and path selection. A shorter AS Path is generally preferred. Engineers may also prepend AS numbers to influence routing. AS Path is a well-known mandatory attribute. It is one of the most important BGP attributes.

Q7. What is Local Preference?

Local Preference is an attribute used inside an AS to influence outbound traffic. Higher Local Preference is preferred. It is propagated through iBGP but not eBGP. The default value is 100. It is commonly used to select a preferred ISP. Changes affect all routers within the AS.

Q8. What is MED?

Multi Exit Discriminator influences how another AS sends traffic into your AS. Lower MED values are preferred. It is exchanged between neighboring ASes. It is often used when multiple links exist between organizations. MED is sometimes called the BGP metric. It helps influence inbound traffic patterns.

Q9. What is Next-Hop Attribute?

The Next-Hop attribute identifies the IP address used to reach a destination. In iBGP environments, next-hop issues are common. The next-hop-self command is frequently used to solve reachability problems. The next-hop must exist in the routing table. If unreachable, the BGP route cannot be installed.

Q10. What is Route Reflector?

A Route Reflector reduces the need for a full-mesh iBGP topology. Clients send routes to the reflector, which redistributes them to other clients. This improves scalability in large networks. Route reflectors simplify administration. They are widely used in ISP and enterprise networks.

Q11. Why is iBGP full mesh required?

By default, iBGP-learned routes are not advertised to another iBGP peer. Therefore, all iBGP routers must form peer relationships with each other. Route Reflectors or Confederations solve this limitation. This behavior prevents routing loops. Large environments rarely use a complete mesh due to scalability concerns.

Q12. What is BGP Synchronization?

Synchronization was an older rule requiring IGP awareness before advertising BGP routes. Modern networks usually disable synchronization. It is rarely used today due to widespread iBGP deployments. The feature was important when transit routers did not run BGP. Modern best practices no longer require it.

Q13. What is BGP Convergence?

Convergence is the time required for BGP to learn and install the best routes after a change. BGP convergence is slower than OSPF or EIGRP because it prioritizes stability and scalability. Route flapping can impact convergence. Various tuning techniques can improve convergence times.

Q14. What is BGP Peering?

BGP peering is the neighbor relationship established between routers. Peers exchange routing information over TCP port 179. Successful peering requires reachability and matching configurations. Neighbor states can be verified using show commands. Authentication can also be configured between peers.

Q15. How does BGP select the best path?

BGP evaluates attributes in order such as Weight, Local Preference, locally originated routes, AS Path length, Origin, MED, eBGP over iBGP, and IGP metric. The route with the most preferred attributes becomes best. Only the best route is normally installed in the routing table. Understanding path selection is critical for troubleshooting.

Q16. What is AS Path Prepending?

AS Path Prepending artificially adds extra AS numbers to a route advertisement. This makes the path appear longer and less attractive. It is commonly used for inbound traffic engineering. It influences decisions made by external ASes. It is one of the most common BGP policy techniques.

Q17. What is Community Attribute?

Communities are tags applied to BGP routes. They simplify policy management by grouping routes logically. ISPs often use communities for route control and traffic engineering. Communities reduce configuration complexity. Well-known communities such as no-export are frequently used.

Q18. Which commands are commonly used to troubleshoot BGP?

Useful commands include show ip bgp, show ip bgp summary, show ip bgp neighbors, show ip route bgp, and debug ip bgp. These commands help verify peers, routes, and path selection. They also help validate received and advertised routes. Troubleshooting should always start with neighbor status verification.

Q19. What are the BGP Neighbor States?

Answer:

BGP forms neighbor relationships through a series of states before reaching the Established state. These states help engineers identify where peering is failing. Understanding these states is essential for troubleshooting BGP synchronization problems. The final goal is to reach the Established state where routes are exchanged.

BGP States

1. Idle

- Initial BGP state.
- Router is waiting to start the BGP process.
- Common issues include incorrect neighbor IP or configuration errors.

2. Connect

- Router attempts to establish a TCP session on port 179.
- Successful TCP connection moves to OpenSent.
- Failure returns the router to Active state.

3. Active

- Router is actively trying to establish a TCP connection.
- Usually indicates reachability or firewall issues.
- A common troubleshooting state in production environments.

4. OpenSent

- TCP session is established.
- BGP OPEN message has been sent to the neighbor.
- Router is waiting for an OPEN message in return.

5. OpenConfirm

- OPEN messages have been successfully exchanged.
- Routers exchange KEEPALIVE messages.
- Successful KEEPALIVE moves the session to Established.

6. Established

- BGP peering is fully operational.
- Routes are exchanged between neighbors.
- This is the desired and stable operational state.

BGP Troubleshooting Commands

show ip bgp summary - Verify BGP neighbor status, prefixes received and uptime.

show ip bgp - View entire BGP table and best-path selection.

show ip bgp neighbors - Check neighbor details, capabilities and timers.

show ip route bgp - Verify BGP routes installed in routing table.

show ip bgp neighbors x.x.x.x advertised-routes - Check routes being advertised to a peer.

show ip bgp neighbors x.x.x.x received-routes - Verify routes received from a neighbor.

show ip cef - Validate forwarding information and next-hop resolution.

show ip route - Verify next-hop reachability and route preference.

ping - Test connectivity to neighbor and next-hop addresses.

traceroute - Identify where traffic is failing along the path.

show tcp brief - Confirm TCP 179 session establishment.

show run | section bgp - Review active BGP configuration quickly.

debug ip bgp - Real-time BGP troubleshooting and route updates.

Recommended BGP Troubleshooting Flow

1. Verify IP reachability using ping and traceroute.
2. Check BGP neighbor status using show ip bgp summary.
3. Verify routes are received in the BGP table.
4. Verify routes are installed in the routing table.
5. Check next-hop reachability.
6. Verify route-maps, prefix-lists and filters.
7. Validate BGP attributes such as Weight, Local Preference and AS Path.
8. Perform end-to-end application testing.

Your BGP neighbor is stuck in Active state. What would you check?

Answer:

I would verify IP reachability using ping and traceroute, check that TCP port 179 is not blocked, confirm neighbor IP and ASN configurations, verify routing to the neighbor, and review ACLs or firewalls that may be preventing the TCP session from being established.